Math 526, F08
Quiz 4

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (8 points) Which of the following subsets of $\mathbb{R}^3$ are actually subspaces? (no explanation needed)

(a) The plane of vectors (b_1, b_2, b_3) with $b_1 = b_2$.

Yes

(b) The plane of vectors (b_1, b_2, b_3) with $b_1 = 1$.

No

(c) All linear combinations of $\mathbf{v} = (1, 4, 0)$ and $\mathbf{w} = (2, 2, 2)$.

Yes

(d) All vectors that satisfy $b_1 + b_2 + b_3 = 0$.

Yes

Problem 2. (12 points) Let $\mathbf{A} = \begin{bmatrix} -1 & 1 & -3 \\ 2 & -4 & 1 \\ -3 & -3 & -4 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} -6 \\ 4 \\ -2 \end{bmatrix}$.

(a) Find the **LU** factorization of **A**.

Solution:

$$A = \begin{bmatrix} -1 & 1 & -3 \\ 2 & -4 & 1 \\ -3 & -3 & -4 \end{bmatrix} \xrightarrow{\ell_{21}=-2} \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ -3 & -3 & -4 \end{bmatrix} \xrightarrow{\ell_{31}=3} \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & -6 & 5 \end{bmatrix}$$

$$\xrightarrow{\ell_{32}=3} \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & 0 & 20 \end{bmatrix} = U$$

$$\Rightarrow A = LU$$

$$= \underbrace{\begin{bmatrix} 1 & & \\ -2 & 1 & \\ 3 & 3 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & 0 & 20 \end{bmatrix}}_{U}$$

(b) Using the above factorization to solve $\mathbf{Ax} = \mathbf{b}$ by solution of two triangular systems $\mathbf{Lc} = \mathbf{b}$ and $\mathbf{Ux} = \mathbf{c}$.

Solution: $$Lc = b \Rightarrow \begin{bmatrix} 1 & & \\ -2 & 1 & \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} -6 \\ 4 \\ -2 \end{bmatrix} \Rightarrow \begin{array}{r} c_1 = -6 \\ -2c_1 + c_2 = 4 \\ 3c_1 + 3c_2 + c_3 = -2 \end{array}$$

$$\Rightarrow \begin{array}{l} c_1 = -6 \\ c_2 = 4 + 2c_1 = -8 \\ c_3 = -2 - 3c_1 - 3c_2 = 40 \end{array}$$

$$Ux = c \Rightarrow \begin{bmatrix} -1 & 1 & -3 \\ & -2 & -5 \\ & & 20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -6 \\ -8 \\ 40 \end{bmatrix} \Rightarrow \begin{array}{r} -x_1 + x_2 - 3x_3 = -6 \\ -2x_2 - 5x_3 = -8 \\ 20x_3 = 40 \end{array}$$

$$\Rightarrow \left.\begin{array}{l} x_3 = 2 \\ -2x_2 = -8 + 5x_3 \Rightarrow x_2 = -1 \\ x_1 = 6 + x_2 - 3x_3 = -1 \end{array}\right\} \Rightarrow x = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$$